LifeLink

A Secure Infrastructure-Independent Emergency Communication System

D1 Project Proposal and Initial Literature Review

TED University | Department of Computer Engineering | CMPE 491
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Project information	Details
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Project description

LifeLink is a proposed emergency communication system for situations in which mobile networks and internet access are unavailable. Earthquake rescue is the primary design and evaluation scenario. The system combines a low-power wearable for help requests and pulse indications, a separate offline messaging channel, and a local gateway with a rescue-team dashboard. Communication relies on the project's powered local equipment and an available radio path. The messaging interface and radio technology will be selected by the team after feasibility and cost checks.

The wearable will attempt pulse measurements after manual or rule-based motion activation and report whether the measurement is usable or uncertain. A help request will not depend on successful pulse measurement. The gateway will verify messages, manage excessive traffic, display measurement freshness, and relay an acknowledgement actually issued by a rescue operator. A small machine-learning model will be investigated for measurement-quality assessment and later on-device use as TinyML. LifeLink will support independent rescue work; it will not diagnose a person's condition or interpret silence as evidence that no living person is present.

Proposal objective

We aim to develop and evaluate a small end-to-end prototype that combines help communication, uncertainty-aware pulse reporting and bounded traffic control. The project brings together embedded systems, cybersecurity and machine learning. This proposal defines the intended work; implementation, range, battery life and rescue effectiveness have not yet been demonstrated.

1 Problem and proposed system

Problem and intended users

A person affected by an earthquake may be unable to speak, operate a phone or press a help button. Someone who can communicate may need to describe a situation and receive a reply when normal services are unavailable. LifeLink addresses these two access needs through automatic wearable reporting and a separate messaging channel. It requires local receivers and sufficient energy; wearing the device alone does not establish a communication service.

Component	Planned responsibility
Wearable	Manual help button, motion-based activation, periodic optical pulse acquisition, measurement-quality status, device-presence reports and user feedback. A removable capsule and interchangeable soft wrist strap are the initial form concept.
Offline messaging channel	Short help messages and operator replies without mobile internet. A local Wi-Fi browser page and a radio-based interface remain alternatives for team selection; neither is presented as a finalized implementation.
Gateway and dashboard	Collect and verify messages; show wearable and messaging sessions; distinguish request, receipt and operator acknowledgement; display measurement age and last contact; apply bounded traffic controls.
Field relay nodes	Possible second-semester extension for collection and forwarding. They are not required for the initial direct-to-gateway demonstration.

Core interaction

The wearable wakes through a help-button action or a controlled motion trigger. It attempts a pulse measurement and sends a quality-labelled report. If pulse acquisition fails, it can still report device presence. An operator can acknowledge a help request through the dashboard. The wearable indicates that acknowledgement using vibration, with an LED as a secondary channel. Local request creation or radio receipt must not be presented as operator acknowledgement.

In the messaging flow, a person sends a short message through the selected local interface, the message appears on the shared dashboard, and an operator reply returns through that interface. If the Wi-Fi option is selected, a gateway-hosted browser page is the proposed first demonstration; no installed phone application or field node would be needed for that option.

Design boundaries

Initial reporting distinguishes pulse indication, device detected with uncertain measurement, and missing expected response from a previously observed device. Planned silence during a confirmed restriction is shown separately. Device identifiers do not establish coordinates. Precise localization, respiration radar, room imaging, medical diagnosis and rescue-priority decisions are outside the core scope. People without devices or an available link must still be searched for independently.

2 Initial literature review

This initial review connects research on disaster communication and optical sensing with an existing off-grid platform and an authenticated-encryption standard. It is a focused starting review, not an exhaustive survey. The following implications for LifeLink are our design conclusions; they are not results reported for our prototype.

2.1 Communication after an earthquake

Pueyo-Centelles and colleagues propose a LoRa-based system through which citizens can report their status to emergency organizations using predefined messages and a simple interaction [1]. This is directly relevant to LifeLink's help-request path: short, structured messages can be considered alongside richer user messages. The paper establishes related prior work; disaster messaging over LoRa is not a new contribution by itself.

For LifeLink, radio selection must also consider receive windows, delayed operator acknowledgements, airtime and the energy cost of listening. An available radio technology is not equivalent to a complete delivery protocol. We will assess these requirements in our own prototype and will not transfer published communication results to rubble conditions without testing.

2.2 An existing off-grid platform

Meshtastic is an open-source platform that uses LoRa radios for off-grid mesh communication and text exchange without relying on existing cellular or internet infrastructure [2]. It provides a practical reference for local messaging and radio integration. We have not committed to adopting Meshtastic or its protocol.

LifeLink's proposed evaluation will additionally examine pulse-report quality, measurement freshness, genuine operator acknowledgement and traffic restriction that preserves bounded help access. These are our selected integration requirements, not a claim that all related systems lack them. A fuller comparison of candidate platforms and their delivery semantics will be part of D2.

2.3 Wrist PPG and reference data

Reiss and colleagues present Deep PPG, a convolutional approach to heart-rate estimation, and introduce PPG-DaLiA [3]. The dataset provides recordings from 15 participants during daily activities; its wrist PPG and acceleration are paired with an ECG-based reference [4]. It offers a suitable starting point for studying movement-related errors before collecting our own data.

LifeLink's initial model will have a different target from direct heart-rate regression: it will estimate whether a fixed, simple pulse extractor is within a stated error tolerance for a window. The ECG reference will be used for labels and evaluation only. Participants will be kept separate across training and test. PPG-DaLiA is not an earthquake or rubble dataset; its sensor and conditions do not establish performance for our selected optical module.

2.4 Body carried rescue transmitters

Avalanche transceivers provide a commercial example of body-carried rescue signalling. Mammut's Barryvox documentation describes an upper-body carrying system and a 457 kHz transmitter [7]. This motivates examining wearability and readiness as well as communication. Its avalanche search performance does not establish operation through building debris; LifeLink will evaluate its own pulse-reporting and messaging functions separately.

2 Initial literature review continued

2.5 Survivor detection beneath rubble

Rescue teams also use methods that do not require a transmitter on the affected person. NASA JPL describes FINDER radar sensing small motions associated with breathing and heartbeat beneath rubble [8]; Savox describes Delsar seismic/acoustic listening for trapped victims [9]. LifeLink instead receives reports from a carried device and supports help communication. It complements these search methods: device-free people and areas without a usable link still require independent search. Published radar or acoustic performance cannot be transferred to our radio link.

2.6 Signal quality features

Elgendi investigates several PPG signal-quality indices, including skewness and relative power [5]. This motivates examining inexpensive shape and spectral features rather than beginning with a large neural network. Published quality-index findings do not directly establish reliability on a different wrist sensor or under our error-based label definition.

Our initial comparison will use rules, logistic regression and a small multilayer perceptron. Candidate features describe PPG peak power, a competing peak, motion variability, PPG–motion frequency separation and waveform skewness. The learned models will be judged against the same fixed pulse extractor and reference labels. Complexity will be retained only if it provides useful error–coverage performance.

2.7 Authentication and confidentiality

NIST SP 800-38C defines CCM, which combines confidentiality and message authentication for a symmetric block cipher [6]. AES-CCM is therefore a candidate for protecting the wearable's reports and commands. Selecting it would still require a concrete design for keys, unique nonces, counters, associated data and restart handling; encryption alone does not supply replay protection or traffic fairness.

The proposal uses temporary device identifiers without requiring personal identity. They must be authenticated, and the gateway may still link successive reports from a device. We do not claim unlinkability from an authorized gateway. Personal medical profiles and continuous location tracking are not required for the first prototype.

Proposed contribution and review limits

Our intended contribution is the integration and evaluation of an offline help channel, quality-labelled pulse indications, truthful acknowledgement states and bounded traffic control in one small system. We do not claim to be the first disaster radio, wearable pulse monitor or encrypted messaging system. The key question is whether these functions can operate together within the selected hardware, energy and airtime constraints.

Literature direction	How it informs LifeLink
Earthquake status messaging [1]	Predefined help reports and a simple civilian interaction.
Off-grid mesh platform [2]	A practical communication reference; adoption remains open.
Body-carried rescue transmitter [7]	A commercial reference for wearing and readiness; different propagation conditions.
Deep PPG and PPG-DaLiA [3, 4]	Motion-aware data and ECG reference for participant-separated evaluation.
PPG quality indices [5]	Small feature-based models and explicit quality baselines.
CCM standard [6]	A candidate basis for report and command protection.

3 Scope work plan and evaluation

Period	Planned output
Semester one	Requirements, analysis and high-level design reports; a small wearable–gateway flow and a separate offline messaging flow on one dashboard; rule-based measurement quality; authenticated temporary identifiers; a bounded routine-report restriction; an initial computer-based quality model.
Semester two	Deploy the feasible model on the chosen MCU; measure the full processing pipeline; expand range/obstruction and multi-device load tests; investigate automatic traffic control and field relays if feasible.

Initial demonstration

Flow A: controlled activation → pulse or uncertain measurement and device report → dashboard → actual operator acknowledgement → wearable feedback. A timed restriction pauses routine reports, while bounded manual help and scheduled control listening remain available; routine reporting resumes after expiry. Flow B: a short help message reaches the same dashboard and an operator reply returns through the selected messaging interface. Internet access will be disabled during the demonstration. Channel-specific details will be finalized with the team.

Evaluation approach

Communication tests will record message delivery, delay and acknowledgement status. Traffic-quarantine tests will compare the same load with the mechanism enabled and disabled, including delayed/lost help messages and false restrictions. Authentication tests will include modified or replayed commands. Power tests will measure the actual board rather than infer whole-device current from chip data sheets.

For quality assessment, a fixed spectral pulse extractor will be evaluated against ECG-derived reference labels. Models and rules will be compared at similar coverage, reporting errors among accepted windows and results by participant and activity. The initial model candidates are logistic regression and a small single-hidden-layer MLP. Model deployment, memory, latency and energy measurement belong to semester two; improvement over rules is not assumed.

Feasibility and responsibilities

MAX30101 and a low-power accelerometer are sensor candidates. MCU, radio, battery chemistry and capacity remain subject to laboratory inventory, staged purchasing and measured feasibility. No specific range or multi-year service life is promised. Wearable comfort, battery protection and maintenance are design constraints. Human measurements will follow the applicable university ethics process; open data and non-human hardware tests allow work to continue meanwhile.

The embedded-systems role covers sensing, activation, power and feedback; cybersecurity covers authentication, temporary identifiers and traffic control; AI covers quality rules, labels and model evaluation. Integration, messaging-interface selection and reports are shared work. Individual task assignment will be agreed by the team.

4 Milestones and references

Deadline	Deliverable
16 October 2026	D1 Project proposal and initial literature review.
23 October 2026	D2 Project specifications and literature review; D3 backlog.
20 November 2026	D4 Analysis report; D5 updated backlog.
31 December 2026	D6 High-level design report; D7 updated backlog.
4–17 January 2027 TBA	D8 Presentation, poster and initial demonstration.

Dates follow the semester-specific schedule in the CMPE 491 syllabus. Final interface, radio and component choices will be recorded as decisions, rather than treated as assumptions already agreed by the team.

References

[1] R. Pueyo-Centelles et al., “A LoRa-Based Communication System for Coordinated Response in an Earthquake Aftermath,” Proceedings, vol. 31, no. 1, art. 73, 2019. doi: 10.3390/proceedings2019031073. https://www.mdpi.com/2504-3900/31/1/73

[2] Meshtastic, “Introduction,” official project documentation. https://meshtastic.org/docs/introduction/ (accessed 15 October 2026).

[3] A. Reiss, I. Indlekofer, P. Schmidt, and K. Van Laerhoven, “Deep PPG: Large-Scale Heart Rate Estimation with Convolutional Neural Networks,” Sensors, vol. 19, no. 14, art. 3079, 2019. doi: 10.3390/s19143079. https://www.mdpi.com/1424-8220/19/14/3079

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[5] M. Elgendi, “Optimal Signal Quality Index for Photoplethysmogram Signals,” Bioengineering, vol. 3, no. 4, art. 21, 2016. doi: 10.3390/bioengineering3040021. https://www.mdpi.com/2306-5354/3/4/21

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